Name: M. Moiz Ansari   Roll no: 23i-0523
Section: F

Assignment-3

Q1: (a)

4 IPs :

1. 10111001 · 00100010 · 00110100 · 00000000
2. 10111001 · 00100010 · 00110100 · 01000000
3. 10111001 · 00100010 · 00110100 · 10000000
4. $\underbrace{10111001 \cdot 00100010 \cdot 0011010}_{N}\underbrace{1 \cdot 00000000}_{H}$

There is unique combination till 23 bits. So best smallest aggregate block is smallest of these with first 23 common bits.

⇒ firsone: 185.34.52.0/23 Ans.

(b)

Another IP: 198.51.120.128/26

⇒ $\underbrace{11000110 \cdot 00110011 \cdot 01111000 \cdot 10}_{N}\underbrace{000000}_{H}$

198.51.120.128 → Network IP
198.51.120.191 → Broadcast IP
⇒ $\boxed{198.51.120.130}$ → Random host IP
Ans.

(c)

ISP Block → 198.51.120.0/24

For dividing into 5 subnets, it requires atleast 3 bits in network part from host. So we borrow 3 bits from host.

11000110 · 00110011 · 01111000 · 00000000
" · " · " · 001 "
" · " · " · 010 "
" · " · " · 111 "
" · " · " · 100 "

⇒ a . b . c . d / x

198 . 51 . 120 . 0 / 27

198 . 51 . 120 . 32 / 27

198 . 51 . 120 . 64 / 27

198 . 51 . 120 . 96 / 27

198 . 51 . 120 . 128 / 27